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Performance Analysis and Optimization of Finite Impulse Response Filters Using Allan Variance ⋆

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Abstract: The design of filters seeks a separation of noise from a desired signal, and the boundary between both is a tradeoff that is a fundamental topic in signal theory. In the presence of signals wherein noise properties have time-varying components, this tradeoff is particularly challenging to optimize. The Mean Squared Error (MSE) is a standard performance metric for evaluating the performance of filters. For signals with non-white noise characteristics - which encompass nearly all real-world signals - the calculation of MSE typically requires repeated analysis across multiple experiments. Prior work by the authors introduced and extended Allan VARiance (AVAR) methods, which analyze variances within increasing data windows, to optimize Moving Average (MA) filters. That work suggested an equivalence between the time-consuming iterative process of using the MSE for filter optimization versus an analysis of the area under an AVAR curve, which can be calculated in one step.
This paper extends the use of the AVAR area method for selecting an optimal Finite Impulse Response (FIR) filter, where optimality is defined as the filter that minimizes the MSE between desired and filtered signals. Prior results are further extended to illustrate that the discrete integration of the AVAR curve yields a performance index that, in one step, generates the MSE-optimal filter for input with drift (random walk) corrupted by white noise. AVAR is compared against the MSE to show that both the performance indices give nearly equivalent optimal FIR filter designs. This AVAR FIR filter optimization is achieved with only one iteration versus hundreds of iterations to optimize filters using MSE calculations.

Keywords: Allan variance, Finite impulse response, Signal processing.

1. INTRODUCTION

Allan VARiance (AVAR) is a technique first developed to study the frequency stability of atomic clocks by Allan (1966) and Barnes (1971) at the National Bureau of Standards, Boulder, Colorado for noise types including white noise phase modulation, flicker noise phase modulation, white noise frequency modulation, and flicker noise frequency modulation. The advantage of AVAR versus standard variance is its invariability to data length. Howe et al. (1981) presented confidence bounds for AVAR estimates and empirical formulae to calculate degrees of freedom of AVAR estimate. Galleani and Tavella developed Dynamic Allan VARiance (DAVAR) to study changing clock behavior in the Galileo satellite system in a series of publications (Galleani and Tavella (2003, 2005); Galleani and Tavella (2009); Galleani (2010, 2011)). AVAR applications include quantifying the stability of clocks (Tetu et al. (1973); Busca et al. (1975)), lasers (Kartaschoff and Barnes (1972); Lerberghe et al. (1978); Hori et al. (1983)), interferometers (Rogers and Moran (1981)), oscillators (Giles et al. (1989)), and the characterization of inertial sensors (Li and Fang (2013); Bhardwaj et al. (2015)). While AVAR methods historically focused on hardware and sensor characterization, these same methods readily extend to signal processing and controller design. For example, one recent development uses AVAR to develop an optimal moving average filter for filtering white noise Haeri et al. (2021).

This paper is motivated by a project requiring real-time signal analysis for autonomous vehicles generating 20,000 measurement data points per kilometer. Fast methods of AVAR-type data variance analysis, or Fast-AVAR (FAVAR) were developed by the authors to enable real-time (less than $0.5s$) estimation of massive quantities of road-friction data (Maddipatla et al. (2021)). This prior work used raw data samples, but the need to extend results to MA, FIR, and IIR filtering were observed, which led to the contributions herein. The results of Maddipatla and Brennan (2024) extended AVAR analysis to MA filters. This work aims to extend the prior AVAR analysis of MA filters toward the general form of FIR filters, namely, a moving-average type filter with non-uniform coefficients.

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To motivate the FIR analysis, the definition and notation of signals is useful. Figure 1 shows the system diagram of an FIR filter. Here, $r(k)$ is the reference/true input, $v(k)$ is the noise, $u(k)$ is the filter input, $y(k)$ is the filtered output, and $e(k)$ is the error. The presence of noise, $v(k)$, will cause the output $y(k)$ to be corrupted by transformed noise included in $v(k)$, namely random walk, white noise, etc. A typical goal of using an FIR filter is to minimize the MSE of the error $e(k)$ (e.g., the L-2 norm of the error). Detailed descriptions and definitions of FIR filtering can be found in nearly all standard signal processing textbooks; for example, see, Williams and Taylor (2006); Schlichthärle (2011); Vaidyanathan (1987).

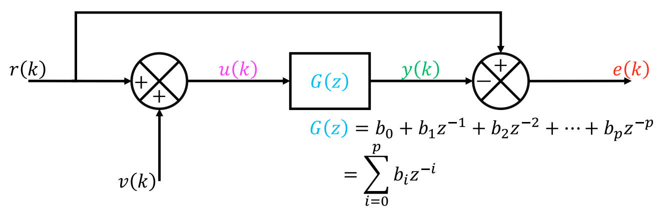

Fig. 1. System diagram of an FIR filter.

This work is organized as follows: Section 2 presents the AVAR extension to FIR filters, comparing analytically-predicted AVAR with the estimated value. Section 3 establishes the area of AVAR as an alternative to MSE to design an optimal FIR filter. Conclusions then summarize the results.

2. FINITE IMPULSE RESPONSE FILTER

This study utilizes FIR filters designed with the Hamming window, as referenced in Harris (1987). The coefficients (b_i) of a p^{th}-order low-pass FIR filter based on the Hamming window are defined by equation (1).

$$b_i = \begin{cases} \dfrac{w_i \sin\left(\pi\omega_n\left(i - 0.5p\right)\right)}{\pi\left(i - 0.5p\right)} & i \neq 0.5p \\ w_i\omega_n & i = 0.5p \end{cases} \tag{1}$$

$$w_i = 0.54 - 0.46\cos\left(2\pi\frac{i}{p}\right) \qquad 0 \leq i \leq p$$

where b_i are filter coefficients, p is the order, and ω_n is the normalized cutoff frequency. The choice of Hamming windows is made mainly due to the ubiquity of this filter design method. The AVAR analysis introduced in this work applies to all other FIR filter design methods as it relies solely on the filter coefficients rather than the design process itself. The output y_j^p of a p^{th}-order FIR filter is given by equation (2).

$$y_j^p = \sum_{i=0}^{p} b_i u_{j-i} \tag{2}$$

where u_{j-i} is the FIR filter input. The normalized cutoff frequency (ω_n) is calculated by dividing the cutoff frequency (ω_c) by the Nyquist frequency (ω_N), as shown in equation (3).

$$\omega_n = \frac{\omega_c}{\omega_N} \tag{3}$$

The choice of initial conditions can significantly influence the behavior of an FIR filter during the initial operation. For an FIR filter of order p, the initial conditions influence only the first p data points of the output signal. Consequently, in the remainder of this work, the final I/O pair is synthesized by excluding the first p data points from the original I/O signals to make them devoid of the effect of initial conditions. Similar to the MA filters, this work uses an input signal of length $N + p$ to synthesize the I/O pair of length N for an FIR filter of order p. An illustration of the input (black dots) and output (circles and asterisks) of an FIR filter with order $p = 50$ and normalized cutoff frequency $\omega_n = 0.25$ is shown in Figure 2. For an input signal of length 70, an FIR filter of order $p = 50$ outputs a signal of length 70. Initial conditions influence the first 50 data points (blue circles) in the output signal, whereas the remaining data (magenta asterisks) are devoid of the effect of initial conditions.

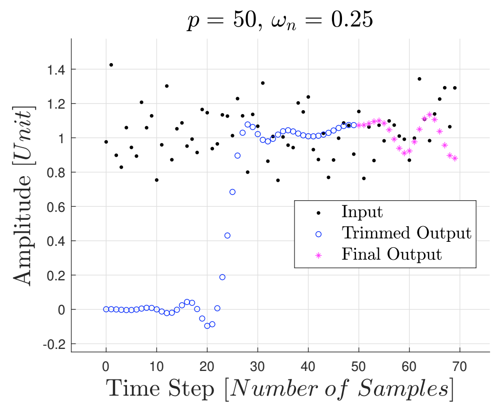

Fig. 2. I/O of an FIR filter with order $p = 50$ and normalized cutoff frequency $\omega_n = 0.25$.

The goal of FIR filtering in this work is to remove the effects of noise without signal loss, resulting in a constant zero error signal; however, in practice, this goal is impossible. For this work, an "under-filtering" FIR filter is defined as one that uses too little data, and an "over-filtering" FIR filter is defined as one that uses too much data. Figure 3*a* (top) illustrates the outputs of under-filtering (brown) and over-filtering (magenta) FIR filters. Figure 3*b* (bottom) shows the corresponding AVAR curves for the errors of the same under-filtering (brown) and over-filtering (magenta) FIR filters. As an introduction to AVAR plots: the x-axis represents how many samples are used to create a data bin, whereas the y-axis represents the variance between averages between all data bins (Maddipatla et al. (2021); Maddipatla and Brennan (2024)). Thus, high-frequency noise content affects variances at smaller correlation intervals and low-frequency noise content is evident at larger correlation intervals. The high-frequency content in the error of an under-filtering FIR filter is evident in Figure 3*b*, which shows higher AVAR at smaller correlation intervals. In contrast, the low-frequency content in the error of an over-filtering FIR filter can be seen as higher AVAR observed at larger correlation intervals. The motivation behind this work is to utilize AVAR to design the best filters; this example illustrates that an optimal FIR filter is one that minimizes AVAR across all correlation intervals, thereby minimizing frequency content due to noise across a wide frequency range.

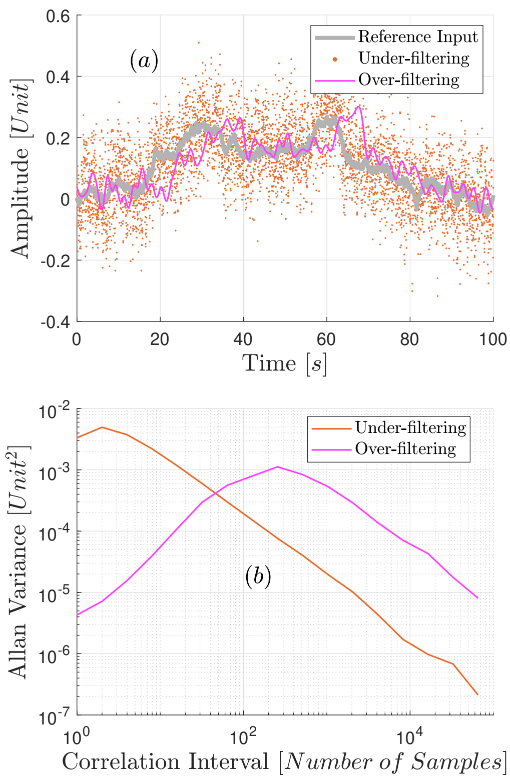

Fig. 3. a) Output of an under-filtering and over-filtering FIR filter. b) AVAR of error in an under-filtering and over-filtering FIR filter.

2.1 Estimated AVAR of FIR filter output

The AVAR estimation of a signal, as presented in Allan (1987), is extended to the output of FIR filters in this work, as shown in equation (4).

$$\sigma_A^2[m] = \frac{1}{2(N-2m)} \sum_{l=2m+1}^{N} \left(\bar{y}_l^p - \bar{y}_{l-m}^p\right)^2 \tag{4}$$

where m is the correlation interval, and N is the data length. Furthermore, $\bar{y}_l^p$ is given by equation (5).

$$\bar{y}_l^p = \frac{1}{m} \sum_{j=l-m}^{l-1} y_j^p \tag{5}$$

where y_j^p is the FIR filter output.

Unless otherwise specified, the reference input signal used in this study is drift (random walk coefficient of 0.02 $\frac{Unit}{\sqrt{s}}$), and the noise is white noise (power spectral density of 0.0004 $\frac{Unit^2}{s}$). To illustrate how FIR filtering affects AVAR curves, Figure 4 shows the AVAR of a low-pass FIR filter's input (black) and output (magenta), along with their respective lower (LB) and upper (UB) 95% confidence bounds (Galleani (2011)). The AVAR of the output is lower than that of the input at shorter correlation intervals because low-pass FIR filters attenuate the high-frequency content, thereby reducing the AVAR at shorter correlation intervals.

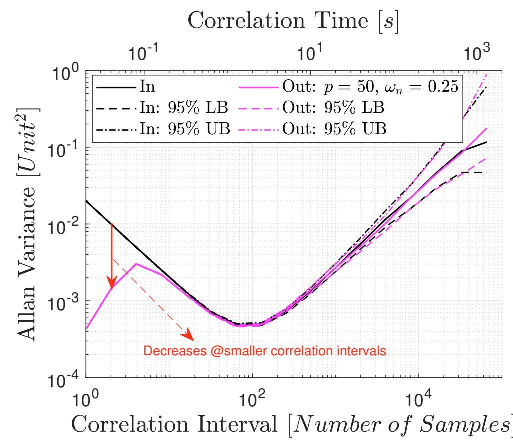

Fig. 4. AVAR of input (In) and output (Out) of an FIR filter with order $p = 50$ and normalized cutoff frequency $\omega_n = 0.25$. In addition to the nominal outputs (solid lines), the 95^{th} percentile lower (LB) and upper (UB) bounds are also shown.

The AVAR of the FIR filter's output for an input signal depends on the filter properties, namely the filter order and the normalized cutoff frequency. Figure 5 illustrates the effect of filter order on the AVAR of the output using low-pass FIR filters with a normalized cutoff frequency of $\omega_n = 0.02$. At higher correlation intervals, there is little to no change in the AVAR despite variations in filter order, as low-pass filters by design do not significantly affect the low-frequency content of the input signal. In contrast, at shorter correlation intervals, the AVAR decreases as the filter order increases. While only low-pass filters are presented here for brevity, the methodology extends directly to high-pass and band-pass filters also.

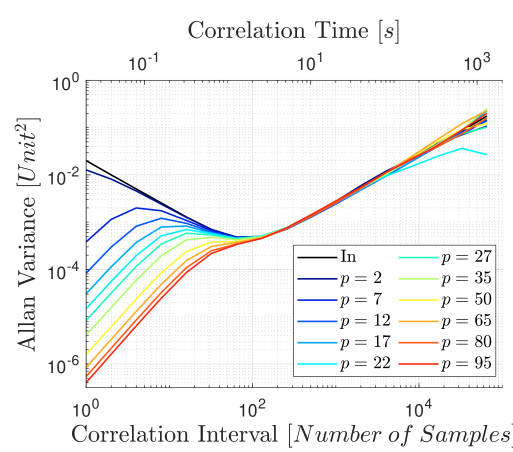

Fig. 5. Effect of filter order p on AVAR of the output of an FIR filter with normalized cutoff frequency $\omega_n = 0.02$.

Figure 6 demonstrates the effect of normalized cutoff frequency on the AVAR of output using low-pass FIR filters with order $p = 50$. Similar to filter order, there is minimal change in AVAR at higher correlation intervals despite variations in normalized cutoff frequency, as low-pass filters do not impact the low-frequency content in the input signal. However, at shorter correlation intervals, AVAR decreases with decreasing normalized cutoff frequency, as expected.

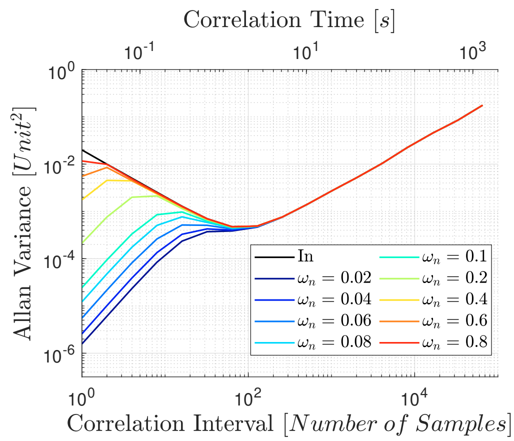

Fig. 6. Effect of normalized cutoff frequency ω_n on AVAR of the output of an FIR filter with order $p = 50$.

2.2 Calculated AVAR of FIR filter output

The AVAR curve of an FIR filter's output can be predicted using the properties of the input signal and the filter coefficients. Equation (6) provides the AVAR of an FIR filter output.

$$\begin{aligned} \sigma_A^2[m] &= \frac{1}{2}\mathbb{E}\left[\left(\bar{y}_l^p - \bar{y}_{l-m}^p\right)^2\right] \\ &= \frac{1}{2}\sum_{i=0}^{p}\sum_{j=0}^{p} b_i b_j \mathbb{E}\left[\Delta\bar{x}_{l-i}\Delta\bar{x}_{l-j}\right] \\ &= \frac{1}{2}\sum_{i=0}^{p} b_i^2 R_\Delta[0] + \sum_{q=1}^{p}\sum_{i=0}^{p-q} b_i b_{i+q} R_\Delta[q] \end{aligned} \tag{6}$$

where $R_\Delta[q]$ is the auto-correlation of 'the difference between successive non-overlapping averages of a signal,' and q is the lag. $R_\Delta[q]$ for white noise and random walk are presented in Greenhall (1991); Galleani (2011).

Figure 7 illustrates that the predicted AVAR (black) calculated using equation (6) falls within the confidence bounds of the estimated AVAR (magenta) obtained from equation (4), indicating that equation (6) effectively predicts the AVAR of FIR output based on filter coefficients and input signal properties. In addition, equation (6) can be utilized to predict the AVAR of errors in MA and FIR filters by treating the errors as the outputs from modified FIR filters.

3. DRIFT INPUT CORRUPTED BY WHITE NOISE

Maddipatla and Brennan (2024) introduced the area under the AVAR curve of error as an efficient alternate measure to MSE for optimizing MA filters. Figure 8 shows that for FIR filters, MSE (dashed lines) is similar to the area under the AVAR curve (solid lines) of error, suggesting that an MSE optimal FIR filter is the same as the filter minimizing area under the AVAR curve. Notably, the area under the AVAR curve offers a computational advantage over MSE in identifying an MSE optimal FIR filter. In this case, the area of AVAR requires only one iteration per filter. In contrast, MSE needs about 500 iterations, and even with this very significant number of iterations and corresponding increase in calculations, the MSE result is less refined than the AVAR method.

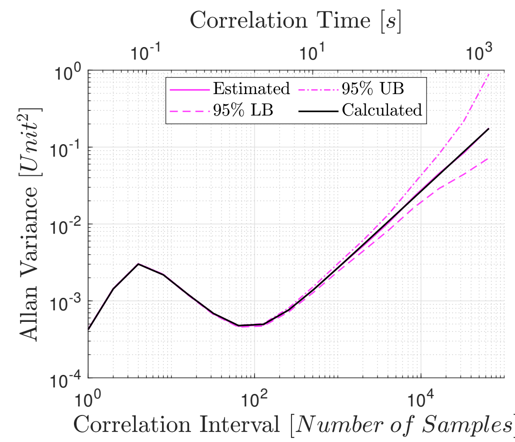

Fig. 7. The calculated AVAR of the FIR filter's output lies within the confidence bounds of the estimated AVAR.

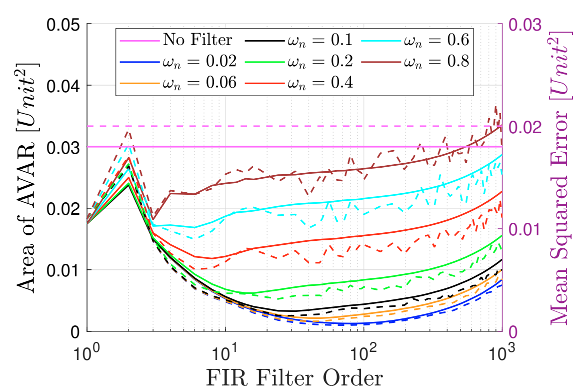

Fig. 8. The error analysis in an FIR filter using the MSE versus the area of AVAR. The MSE (dashed lines) calculation averaged over 500 iterations converges to the same optimal filter as the area of AVAR (solid lines) from 1 iteration.

The FIR filter, which is designed to minimize the area under the AVAR curve of error, aims to produce the best signal in the sense of log-weighted variance across all correlation intervals. Here, the user chooses the normalized cutoff frequency and then estimates the area under the AVAR curve of errors for several filter orders. The optimal FIR filter order is selected to minimize the area under the AVAR curves of errors. Figures 9, 10, and 11 illustrate the selection of the optimal order of FIR filter for various normalized cutoff frequencies and input signals, and shows via blue vertical line the calculated optimal filter order. Figure 9*a* (top) depicts the area under the AVAR curve of error along with MSE. Conversely, Figure 9*b* (bottom) displays the reference input alongside the filter output for FIR filters with a normalized cutoff frequency of 0.02. The 3^{rd}-order filter utilizes too little data, resulting in increased noise in the output (green dots). In contrast, the 548^{th}-order filter utilizes too much data, leading to an increased lag in the output (red line). However, the optimal

80^{th}-order filter output (blue line) effectively balances noise and lag by minimizing the AVAR of error.

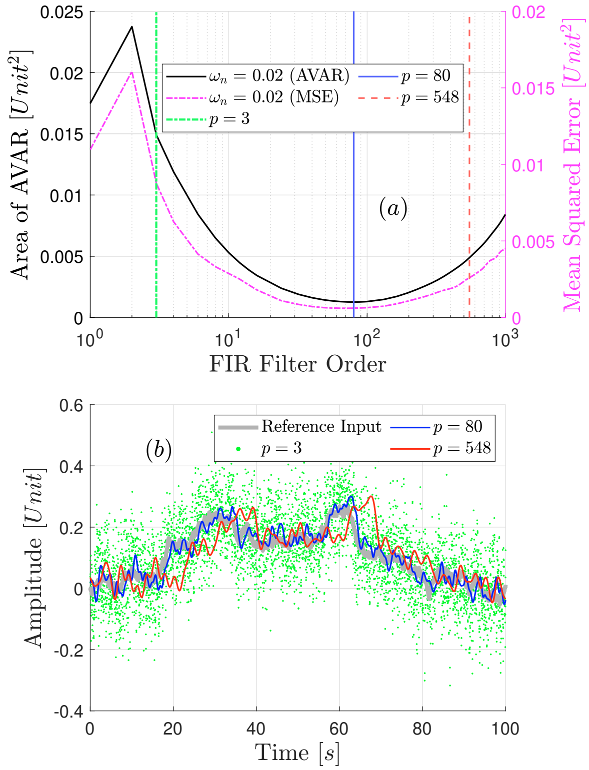

Fig. 9. The optimal filter is the FIR filter with order $p = 80$, minimizing the area of AVAR. $a)$ Area under the AVAR curve of error and MSE in FIR filters with normalized cutoff frequency $\omega_n = 0.02$. $b)$ Output of FIR filter for three different orders.

Similarly, Figure 10 presents FIR analysis using the area under the AVAR curve of error for FIR filters with a normalized cutoff frequency of 0.1. In this particular case, the optimal order of the filter decreases from 80 to 32 as the normalized cutoff frequency increases from 0.02 to 0.1. While the reduction in the optimal filter order with an increased normalized frequency applies to the scenario discussed here, it cannot be generalized. In other words, the optimal order of the filter is not only a function of the normalized cutoff frequency but also dependent on the properties of the reference input and noise.

To show that the filter optimization method does not depend on the specific noise characteristics used earlier, Figure 11 demonstrates FIR analysis for different input signals using the same FIR filters as in Figure 9. Here, the reference input signal is drift (random walk coefficient of 0.02 $\frac{Unit}{\sqrt{s}}$), and the noise is white noise (power spectral density of 0.002 $\frac{Unit^2}{s}$). For this specific choice of input signal and normalized cutoff frequency, the optimal filter order minimizing the area under the AVAR curve of error is 132.

4. CONCLUSIONS

This work presents an AVAR methodology to optimize FIR filter design. Specifically, there are three contributions:

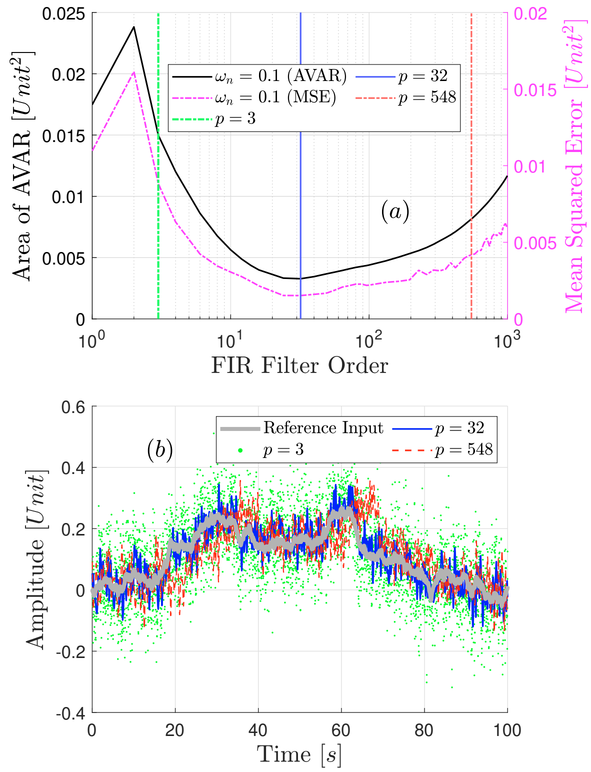

Fig. 10. The optimal filter is the FIR filter with order $p = 32$, minimizing the area of AVAR. $a)$ Area under the AVAR curve of error and MSE in FIR filters with normalized cutoff frequency $\omega_n = 0.1$. $b)$ Output of FIR filter for three different orders.

1) the equations to predict the AVAR curves for FIR filters are presented; these equations are an extension to the ones introduced in Maddipatla and Brennan (2024). Additionally, 2) this work extends the concept of using the area of AVAR to find an MSE-optimal FIR filter, and shows equivalence between MSE-based method and AVAR methods using signals with drift input corrupted by white noise. The main advantages of using AVAR instead of MSE are its applicability to non-white noise signals and the fact that the calculation of area of AVAR needs only one iteration per filter in contrast to many (typically hundreds or more) of iterations required for MSE. Finally, 3) this new optimization method is demonstrated by optimizing an FIR filter's properties for two sample noisy signals. Extending the application of the area of AVAR to signals such as sinusoids and mean-reverting random walk could pose challenges, as the AVAR of such signals has multiple correlation intervals where AVAR is locally minimum.

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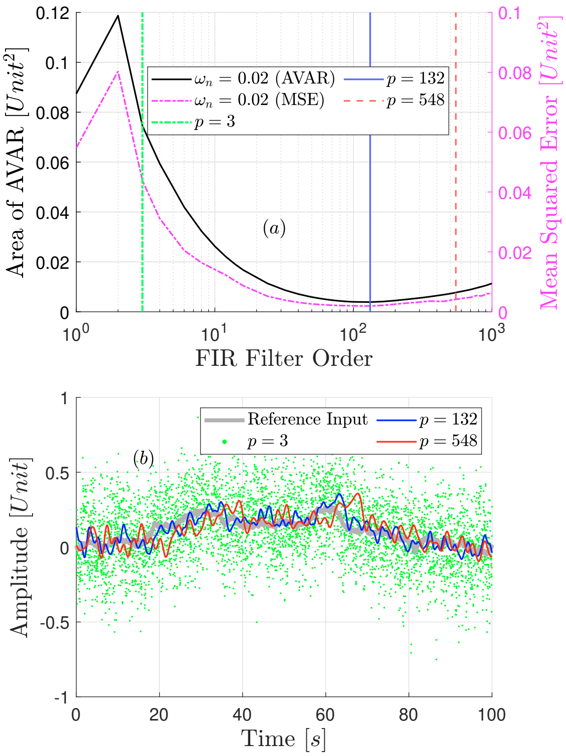

Fig. 11. The optimal filter is the FIR filter with order $p = 132$, minimizing the area of AVAR. a) Area under the AVAR curve of error and MSE in FIR filters with normalized cutoff frequency $\omega_n = 0.02$. b) Output of FIR filter for three different orders.

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